

MS101 – Makerspace
2023-24/I (Autumn Semester)
EE Midsemester Examination

Answers to Numerical Questions

Ques.	Answers	Comments
1	-2.27 V	
2	$f_c = 7957.8\text{ Hz}$, 0.707, 0.2437	
3	a) $V_{\text{DUT}} = 4\text{ V}$, $V_{\text{out}} = -2.667\text{ V}$ b)	
4	a) i) DC ; ii) AC b)	
5	$F = A' B C' + A' B C + A B' C + A B C$	
6	$F = A B D' + A' C' D + B C D' + B' D$	
7	9 bits	
8	1100 : -6 V 1010 : -5 V 0111 : -3.5 V	